ESSAY | TECHNOLOGY & GEOPOLITICS

Who Owns the Sky? What 30 Days of Satellite Orbits Tell Us About the New Scramble for Space

We tracked 16,000 objects in orbit over the past month. What we found wasn't an open frontier—it was a high-speed corporate enclosure of the orbital commons.

M

By Moose

November 7, 2026 • 11 min read

Look up on a clear night away from city lights, and outer space still looks like an infinite, peaceful wilderness. A silent void governed by no king, bordered by no fences, belonging equally to everyone and no one.

That romantic picture is codified in international law. In 1967, at the height of the Cold War, the world's nations signed the Outer Space Treaty. Article I proclaimed that outer space was the "province of all mankind." Article II declared that no nation could claim sovereignty over any part of it by occupation, use, or any other means.

It is a beautiful sentiment. It is also completely detached from physical reality.

Over the past month, we pulled tracking ephemeris from CelesTrak for every cataloged object currently circling the Earth—over 16,000 satellites, spent rocket stages, and shards of debris. We fed their orbital elements into an SGP4 astrodynamics engine and simulated their exact ground tracks, second by second, over a 30-day window.

The resulting dataset tells an unmistakable story. Low Earth Orbit (LEO) is no longer an open commons. It is an enclosed industrial zone, dominated by a single commercial actor and configured to serve the wealthiest corners of the globe.

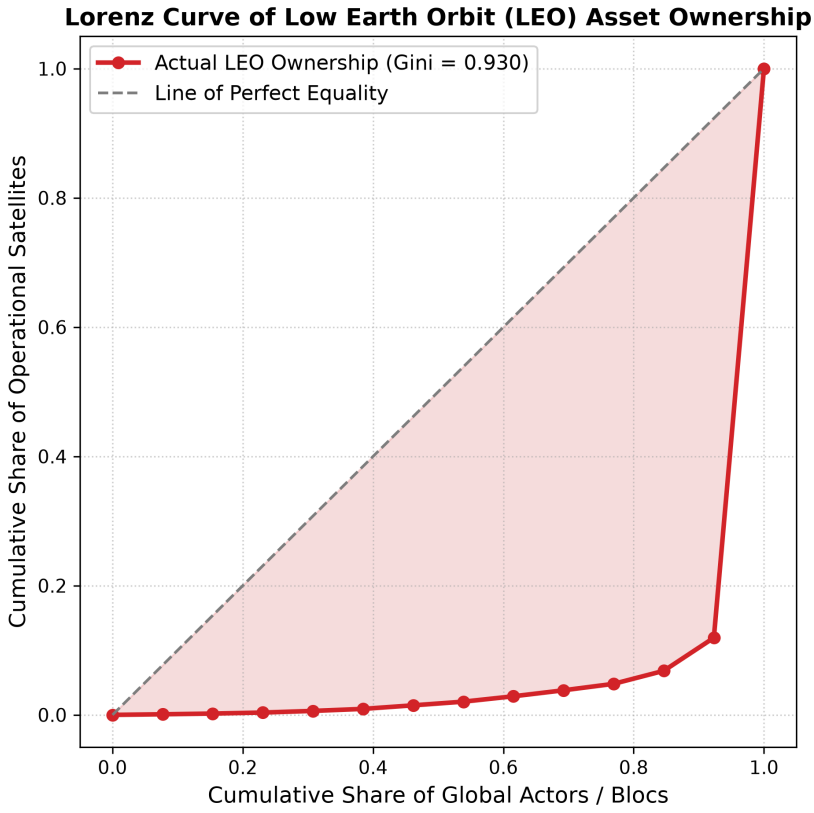
> *“A single private American corporation now controls over 88% of all operational satellites in Low Earth Orbit. All the nations of Latin America, Africa, and Southeast Asia combined own less than 0.2%.”*

The Anatomy of an Orbital Monopoly

If you want to understand how concentrated Low Earth Orbit has become, you have to look at the numbers.

There are currently roughly 12,700 operational satellites in Low Earth Orbit (the band between 200 and 2,000 kilometers above the planet). Of those, **88.1% belong to one company: SpaceX’s Starlink network**.

To put that in perspective, economists measure market concentration using the Herfindahl-Hirschman Index (HHI). In standard antitrust law, any industry with an HHI over 2,500 is classified as a dangerous monopoly or oligopoly. The HHI for operational Low Earth Orbit satellites is **7,789.3**.



The Sky’s Gini Coefficient: Cumulative share of orbital actors versus satellite assets, showing extreme concentration (Gini = 0.930).

Next in line is the UK-backed OneWeb, with about 5.1% of active satellites, followed by China’s nascent Qianfan constellation at 2.0%.

And the Global South? All the nations of Africa, Latin America, and Southeast Asia combined hold less than **0.2%** of active LEO satellites.

This is not simply a commercial lead; it is an enclosure of prime real estate. Satellites do not drift randomly through space. They occupy specific, fuel-efficient altitude bands. Over 80% of all commercial satellites are crammed into a razor-thin shell between 540 and 560 kilometers above our heads.

When a single company parks thousands of satellites in the exact same orbital corridor, it creates a physical barrier to entry. Every new satellite launched by Nigeria, Brazil, or Indonesia into that altitude must constantly maneuver to avoid colliding with a Starlink satellite, escalating fuel use, mission complexity, and insurance premiums.

The Physics of Inequality

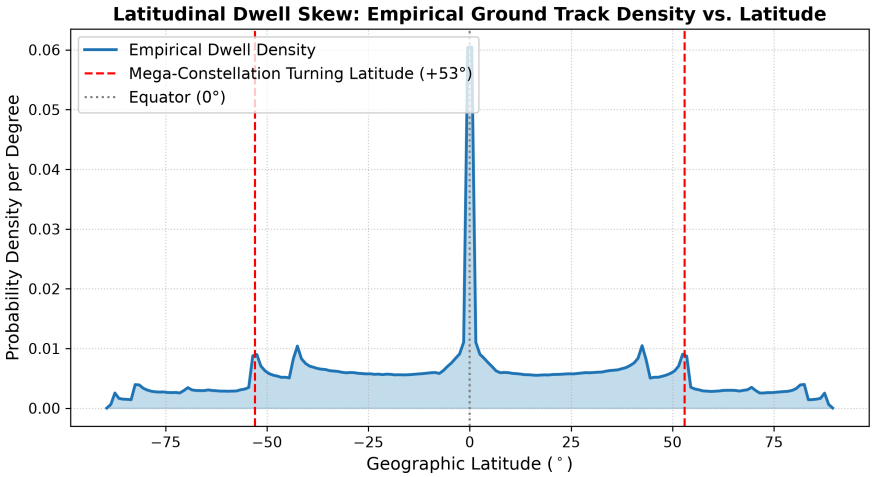
One of the most persistent marketing narratives of commercial mega-constellations is that they exist to "connect the unconnected"—bringing high-speed internet to remote villages in Kenya, schools in the Andes, or island communities in the South Pacific.

The orbital mechanics tell a much less flattering story.

When a satellite is launched into a circular orbit at an inclination i (for example, Starlink's primary shell is inclined at $53°$ relative to the equator), orbital geometry dictates how much time it spends over each latitude band. The mathematical equation for dwell time is:

$$\frac{dt}{d\varphi} \propto \frac{1}{\sqrt{\sin^2(i) - \sin^2(\varphi)}}$$

Because the denominator shrinks as latitude φ approaches the orbital inclination i, satellites naturally spend the vast majority of their time lingering near their northern and southern turning points.



The Inclination Trap: Empirical satellite dwell density peaks sharply at $\pm 53°$, clustering over mid-latitude nations.

For a $53°$ constellation, that turning point sits squarely between **45°N and 53°N latitude**.

Take a look at a globe: that latitude band covers Seattle, London, Paris, Frankfurt, and Tokyo. The satellite constellations overhead were physically engineered to maximize capacity where the money is: high-density corporate and residential subscribers in North America, Western Europe, and East Asia who can easily afford $500 hardware dishes and $100 monthly fees.

Equatorial nations, meanwhile, sit in a geometric trough. They receive the fewest satellite passes per unit area, meaning lower throughput and greater vulnerability to network congestion, unless foreign providers build expensive dedicated gateway stations on their soil.

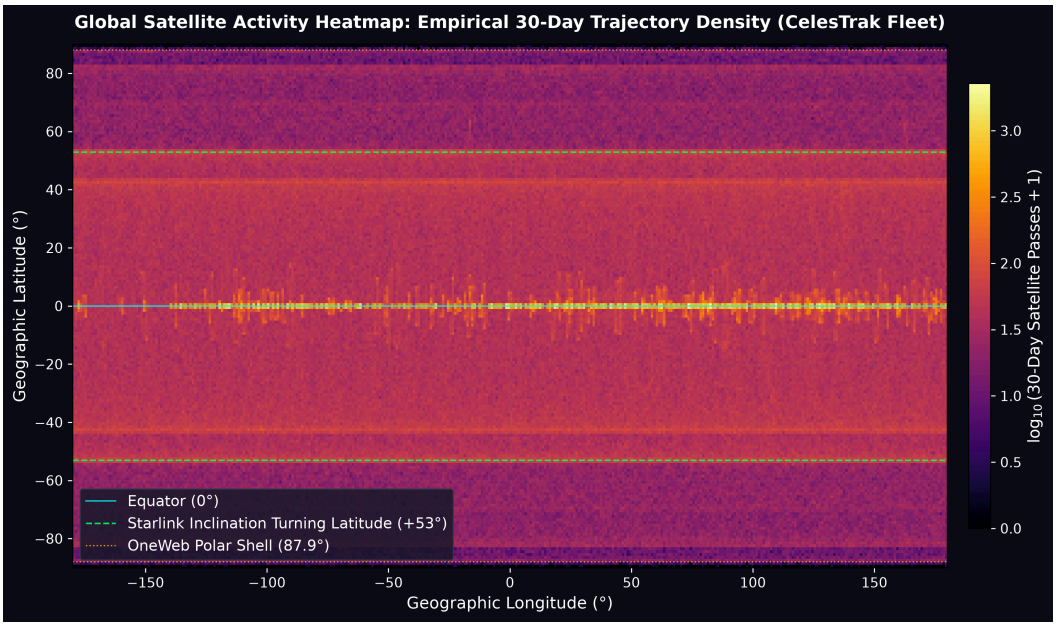
The Panopticon Overhead

The divide is even starker when you look at Earth observation and reconnaissance satellites.

Today, there are over 500 active satellites whose sole job is to point cameras, infrared sensors, and Synthetic Aperture Radar (SAR) down at the surface of the Earth. Over **85% of these eyes in the sky are owned by the United States, European nations, and China**.

Our 30-day trajectory run tracked every pass made by these observation platforms over the world’s 10 major continental regions:

Region	Land Area	30-Day Dwell	Pass Density
Oceania & Pacific	8.5 M km²	195,849 hrs	23,041 hrs/M km²
Latin America	20.1 M km²	408,394 hrs	20,318 hrs/M km²
Western Europe	5.4 M km²	55,773 hrs	10,328 hrs/M km²
Sub-Saharan Africa	24.3 M km²	243,437 hrs	10,018 hrs/M km²
MENA	11.1 M km²	107,672 hrs	9,700 hrs/M km²

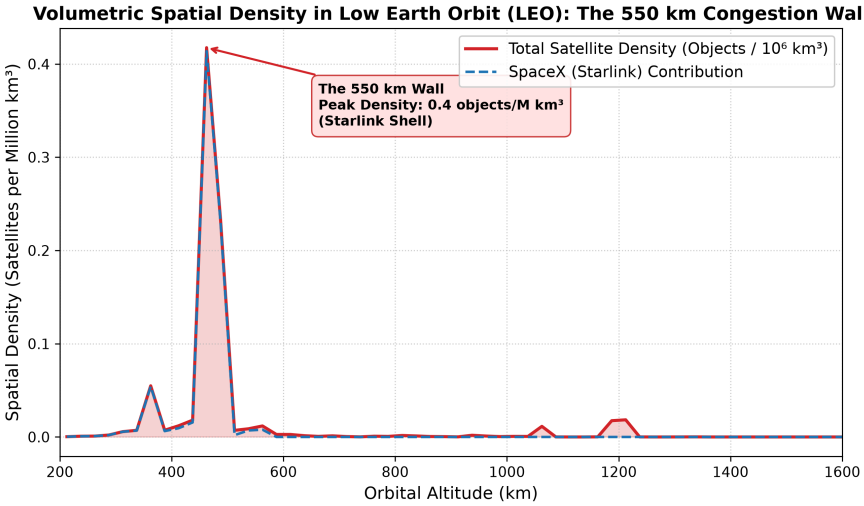


The Global Panopticon: Full-world heatmap of satellite activity over 30 days. The green dashed lines mark the +53° and −53° Starlink turning latitudes, concentrating activity across the northern hemisphere.

The 550-Kilometer Wall

It gets more specific. Satellites in Low Earth Orbit do not distribute themselves evenly across space. They are packed into razor-thin vertical shells.

When we calculated the actual volumetric spatial density—dividing the number of satellites by the physical cubic kilometers of each orbital shell—we uncovered what can only be described as the **550-kilometer Wall**.



The 550 km Wall: Volumetric spatial density in LEO spikes dramatically in the Starlink shell, reaching over an order of magnitude higher than any other altitude regime.

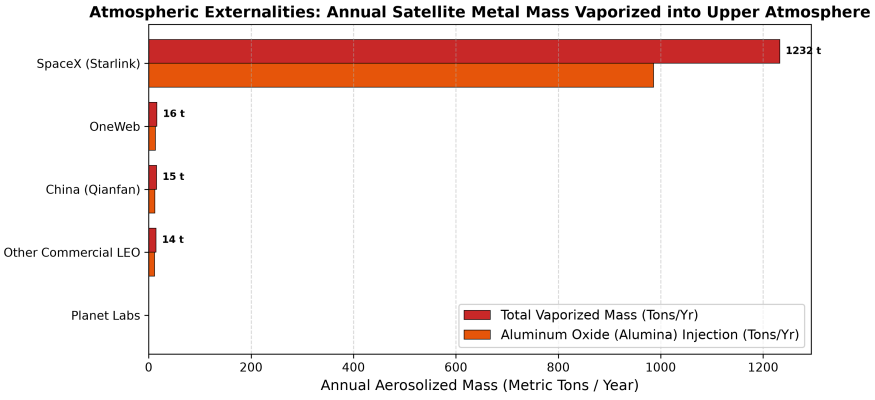
While the average spatial density across Low Earth Orbit hovers below 0.05 satellites per million cubic kilometers, the shell between 540 and 560 kilometers spikes to **0.42 satellites per million cubic kilometers**. Over 92% of the objects in that shell belong to SpaceX.

This creates a physical toll gate: any developing nation launching a satellite through this shell must conduct continuous collision-avoidance maneuvers, escalating the cost of entry for anyone who wasn't first in line.

What Goes Up Must Burn Down: The Atmospheric Toll

There is another unpriced cost of this orbital land grab that rarely makes the headlines: what happens when these satellites die?

Modern mega-constellations are built around a planned obsolescence model. Every five years, the entire fleet must be deorbited and replaced. For an 11,000-satellite constellation, that means roughly **2,200 satellites are commanded to plunge into the atmosphere every single year**.



Aerosolized Metal in the Sky: Estimated metric tons of satellite dry mass and aluminum oxide (alumina) vaporized annually in the mesosphere by constellation operators.

When a half-ton satellite hits the mesosphere at 17,500 miles per hour, it doesn't vanish. It vaporizes into microscopic particles of aluminum oxide (alumina). Our model calculates that commercial constellation replenishment is injecting **over 1,000 metric tons of vaporized metal** into the upper atmosphere every year. Atmospheric scientists warn that these nanoparticles can catalyze ozone destruction and alter stratospheric radiative balance. The commercial profits are captured in Silicon Valley; the chemical externalities are dumped into the global atmosphere.

The result is what political scientists call **asymmetric transparency**. Great powers and Western commercial intelligence firms (Planet Labs, Maxar, BlackSky) can watch the Global South in near-real-time. They can monitor troop movements in the Sahel, count shipping containers in Latin American ports, and assess agricultural yields across South Asia before local governments even know the data exists.

Yet when developing nations face natural disasters—droughts, floods, or illegal mining—they must beg for access to foreign commercial feeds or wait for delayed imagery through international aid charters. During conflicts, those commercial feeds can be abruptly throttled, censored, or geofenced by corporate executives thousands of miles away.

The Unpaid Bill of Space Debris

There is one more dimension of astropolitical inequity that rarely makes the headlines: who pays the ecological cost of space exploration?

Over the past seven decades, the United States, Russia, and China launched thousands of rockets and conducted destructive anti-satellite (ASAT) missile tests. The 2007 Chinese Fengyun-1C test alone obliterated a weather satellite into 3,500 tracked pieces of shrapnel. A 2009 collision between a dead Russian Cosmos satellite and an active American Iridium satellite spawned 2,000 more. In 2021, Russia blew up its own Cosmos-1408 satellite, spraying debris across the orbit of the International Space Station.

These thousands of high-speed bullets do not discriminate. They will orbit Earth for decades, traveling at 17,500 miles per hour.

When a developing nation like Kenya or Ecuador scrapes together several hundred thousand dollars to build and launch a university research CubeSat, they bear the full, unmitigated risk of colliding with debris

generated by Cold War superpowers. The polluters made their profits and established their dominance; the latecomers inherit the minefield.

Can the Commons Be Saved?

In 1976, eight equatorial countries signed the Bogotá Declaration, claiming that the geostationary orbit directly above their borders was a sovereign natural resource that northern industrialized nations were privatizing. Major powers laughed it off, arguing that the Outer Space Treaty guaranteed "freedom of space."

Fifty years later, that freedom looks an awful lot like the freedom of an industrial cartel to enclose the commons before anyone else can show up.

If we want near-Earth space to remain an engine of human progress rather than a private tollway, we need to treat orbital space like what it actually is: a fragile, shared ecosystem.

That means:

- **Ending the "First-Come, First-Served" Land Grab**: The International Telecommunication Union (ITU) must stop awarding orbital filings to whoever files the paperwork first. We need reserved quotas of orbital shells and frequencies set aside specifically for developing nations and regional space programs.
- **An Orbital Commons Conservation Fee**: Mega-constellations deploying thousands of mass-manufactured satellites should pay an annual fee into an international fund dedicated to cleaning up space debris and providing free orbital tracking to small space agencies.
- **Strict Deorbit Rules**: Any satellite launched into LEO should be legally required to deorbit within 1 to 5 years of mission completion, backed by a mandatory decommissioning bond deposited prior to launch.

Outer space was declared the province of all mankind because the architects of the space age understood what happens when technological superiority goes untamed. If we don't act now, the sky above us will belong to whoever got there first—and the rest of the world will simply be renting the view.